

SIMATS ENGINEERING ASSESSMENT TOOL 2

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COURSE NAME: CSA1603

COURSE CODE: Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO3: Evaluate the effectiveness of classification techniques on real-time data for accurate prediction, enabling informed decision-making. (BL3,BL4,BL5)

Assessment Tool Weightage: 30%

QUESTIONS: 3

Total Marks:30

Annexure C

Case study

<i>Criteria</i>	Understanding of Case (2.5)	Application of Concepts (2.5)	Analysis (2)	Solution Design (2)	Clarity (1)	<i>Total(10)</i>
<i>Weightage</i>	25%	25%	20%	20%	10%	<i>100%</i>
<i>Q1</i>						
<i>Q2</i>						
<i>Q3</i>						

Code of Conduct:

I Siva Dakshitha K / 192425395 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Siva Dakshitha

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

ANNEXURE A

Case Study

1. A shepherd boy gets bored tending the town's flock. To have some fun he cries out, "Wolf!" even though no wolf is in sight. The villagers run to protect the flock, but then get really mad when they realize the boy was playing a joke on them.

One night, the shepherd boy sees a real wolf approaching the flock and calls out, "Wolf!" The villagers refuse to be fooled again and stay in their houses. The hungry wolf turns the flock into lamb chops. The town goes hungry. Panic ensues. Identify TP, FP, FN, TN

Q1.

Actual class/ predicted class →	Wolf (Warning) (Yes)	No wolf/ warning (No)	Total
Wolf (Yes)	TP Real wolf came, warning also came	FN No wolf came, But warning is not acted upon.	1
No Wolf (No)	FP No wolf came, warning has come	TN No wolf, No warning	2
Total	3	0	3

- $TP = 1 \rightarrow$ Real wolf came, warning came
 - $FN = 0 \rightarrow$ Real wolf came, but no warning
 - $FP = 2 \rightarrow$ No wolf came, warning came twice
 - $TN = 0 \rightarrow$ No wolf came, no warning
- $\therefore TP = 1, FP = 2, FN = 0, TN = 0$
- $\therefore \text{Total} = 3$

2. Assume there are 100 images, 30 of them depict a cat, the rest do not. A machine learning model predicts the occurrence of a cat in 25 of 30 cat images. It also predicts absence of a cat in 50 of the 70 no cat images. Identify TP, FP, FN, TN and Calculate the Precision and Recall and F1 Score.

Q2.

Actual / predicted	Cat (Yes)	No Cat (No)	Total
Cat (Yes)	TP 25	FN 20	45 30
No Cat (No)	FP 20 5	TN 50	55 70
Total	(45) 30	(55) 70	100

TP, FP, FN, TN

- $TP = 25 \Rightarrow$ cat correctly predicted as cat
- $FN = 5 \Rightarrow$ cat incorrectly predicted as no cat ($30 - 25 = 5$)
- $TN = 50 \Rightarrow$ No cat correctly predicted as no cat
- $FP = 20 \Rightarrow$ No cat incorrectly predicted as cat ($70 - 50 = 20$)

ix Precision

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$= \frac{25}{25 + 20} \Rightarrow \frac{25}{45} \Rightarrow 0.555$$

$$0.555 \times 100 \Rightarrow 55.5\%$$

$$0.833 \times 100 \Rightarrow 83.3\%$$

$$2) \text{ Recall} \Rightarrow \frac{TP}{TP+FN} \Rightarrow \frac{25}{25+20} \Rightarrow \frac{25}{45} \Rightarrow 0.555$$

$$0.555 \times 100 \Rightarrow 55.5\%$$

$$3) \text{ F1-Score} \Rightarrow \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

$$\Rightarrow \frac{2 \times 0.833 \times 0.555}{0.833 + 0.555}$$

$$\Rightarrow \frac{0.92463}{1.388} \Rightarrow 0.6667$$

$$0.6667 \Rightarrow 66.67\%$$

Final Answer

<u>Metric</u>	<u>Value</u>
TP	25
FP	5
FN	20
TN	56

$$\therefore \text{F1-Score} \Rightarrow 66.67\%$$

3. An email service provider wants to classify emails as **Spam or Not Spam** using features:

- Contains “Offer” (Yes/No)
- Contains “Free” (Yes/No)
- Email Length (Short/Long)

Sample Dataset

Offer	Free	Length	Class
Yes	Yes	Short	Spam
Yes	No	Long	Not Spam
No	Yes	Short	Spam
No	No	Long	Not Spam
Yes	Yes	Long	Spam
No	No	Short	Not Spam

Questions

1. Calculate prior probabilities:
 - $P(\text{Spam})$, $P(\text{Not Spam})$
2. Compute conditional probabilities for each feature.
3. Using **Naïve Bayes**, classify:
 - (Offer=Yes, Free=No, Length=Short)
4. Compare results using **WEKA (Naïve Bayes classifier)**.
5. Discuss the **independence assumption** in Naïve Bayes

Q3.

Dataset

Offer	Free	Length	Class
Yes	Yes	Short	Spam
Yes	No	Long	Not spam
No	Yes	Short	Spam
No	No	Long	Not spam
Yes	Yes	Long	Spam
No	No	Short	Not spam

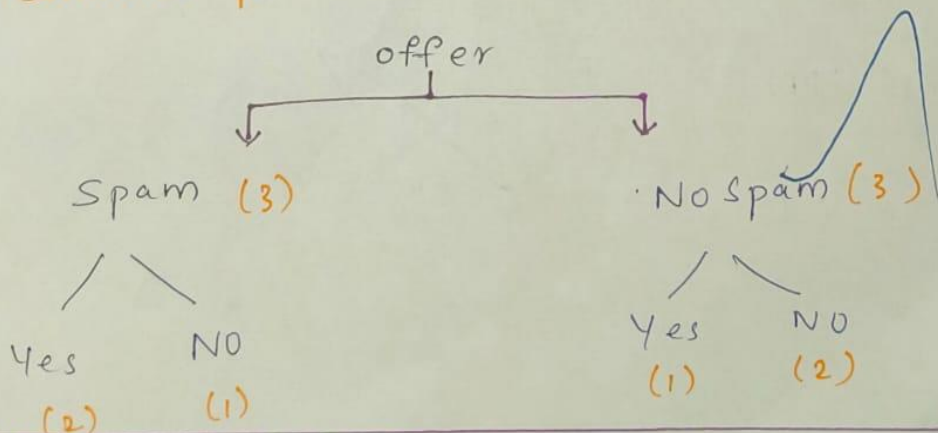
i) Prior Probabilities.

spam $\Rightarrow 3$; No spam = 3 ; Total = 6

$$P(\text{spam}) = \frac{3}{6} = 0.5$$

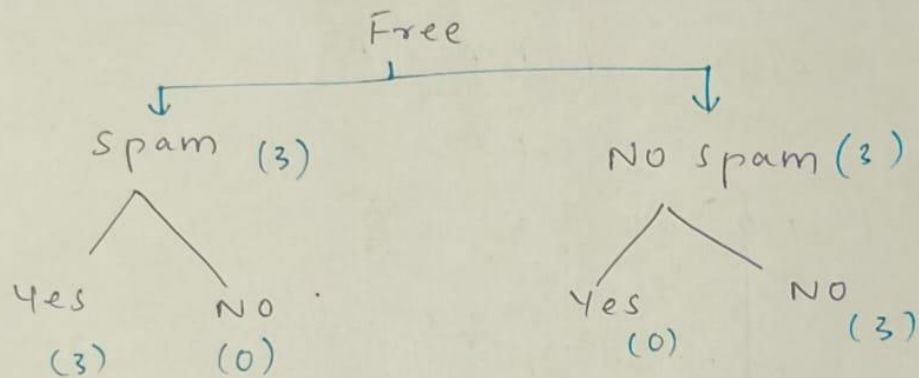
$$P(\text{No spam}) = \frac{3}{6} = 0.5$$

ii) Condition probabilities



$$P(\text{offer} = \text{yes} \mid \text{spam}) = \frac{2}{3} \quad ; \quad P(\text{offer} = \text{yes} \mid \text{no spam}) = \frac{1}{3}$$

$$P(\text{offer} = \text{no} \mid \text{spam}) = \frac{1}{3} \quad ; \quad P(\text{offer} = \text{no} \mid \text{no spam}) = \frac{2}{3}$$

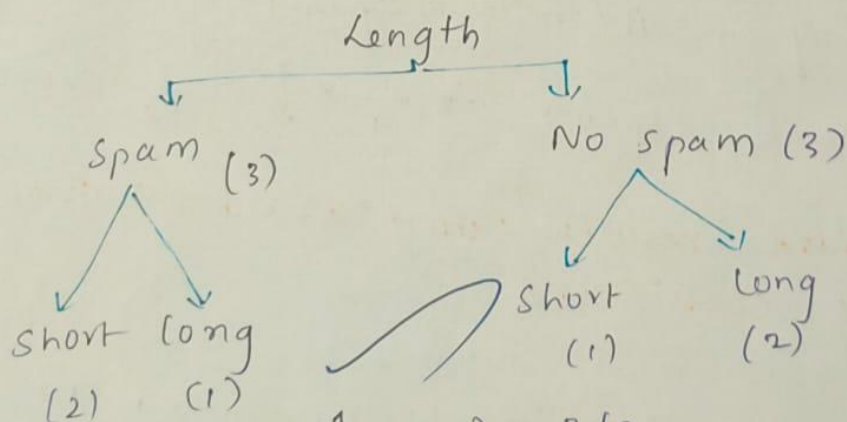


$$P(\text{Free} = \text{yes} \mid \text{spam}) = \frac{3}{3} \Rightarrow 1$$

$$P(\text{Free} = \text{no} \mid \text{spam}) = \frac{0}{3} \Rightarrow 0$$

$$P(\text{Free} = \text{yes} \mid \text{no spam}) = \frac{0}{3} \Rightarrow 0$$

$$P(\text{Free} = \text{no} \mid \text{not spam}) = \frac{3}{3} \Rightarrow 1$$



$$P(\text{length} = \text{Short} \mid \text{spam}) = \frac{2}{3}$$

$$P(\text{length} = \text{Long} \mid \text{spam}) = \frac{1}{3}$$

$$P(\text{length} = \text{Short} \mid \text{not spam}) = \frac{1}{3}$$

$$P(\text{length} = \text{Long} \mid \text{not spam}) = \frac{2}{3}$$

3) Classify the New Email

New Email:

$X = (\text{offer} = \text{Yes}, \text{Free} = \text{NO}, \text{length} = \text{Short})$

Using Naive's Bayes;

$$P(A|B) = \frac{P(B|A) P(A)}{P(B)}$$

$$P(A|\text{mail} = \text{spam}) \Rightarrow \frac{2}{3} \times \frac{0}{3} \times \frac{2}{3} = 0$$

$$\therefore P(A|\text{mail} = \text{spam}) = 0$$

$$P(A|\text{mail} = \text{Nospam}) \Rightarrow \frac{1}{3} \times 1 \times \frac{1}{3} \Rightarrow \frac{1}{9} \Rightarrow 0.111$$

$P(A|\text{mail} = \text{spam}) \times \text{probability of spam}$

$$0 \times \frac{1}{2} \Rightarrow 0$$

probability of $A(\text{mail} = \text{no spam})$

$$P(A|\text{mail} = \text{no spam}) \Rightarrow \frac{1}{9} \times \frac{1}{2} = \frac{1}{18} \Rightarrow 0.0556$$

4) Compare results using Weka (Naive's Bayes classifier)

The dataset was evaluated

The Weka Experiments was used to compare three Bayesian classifiers.

1) BayesNet

2) Naive Bayes

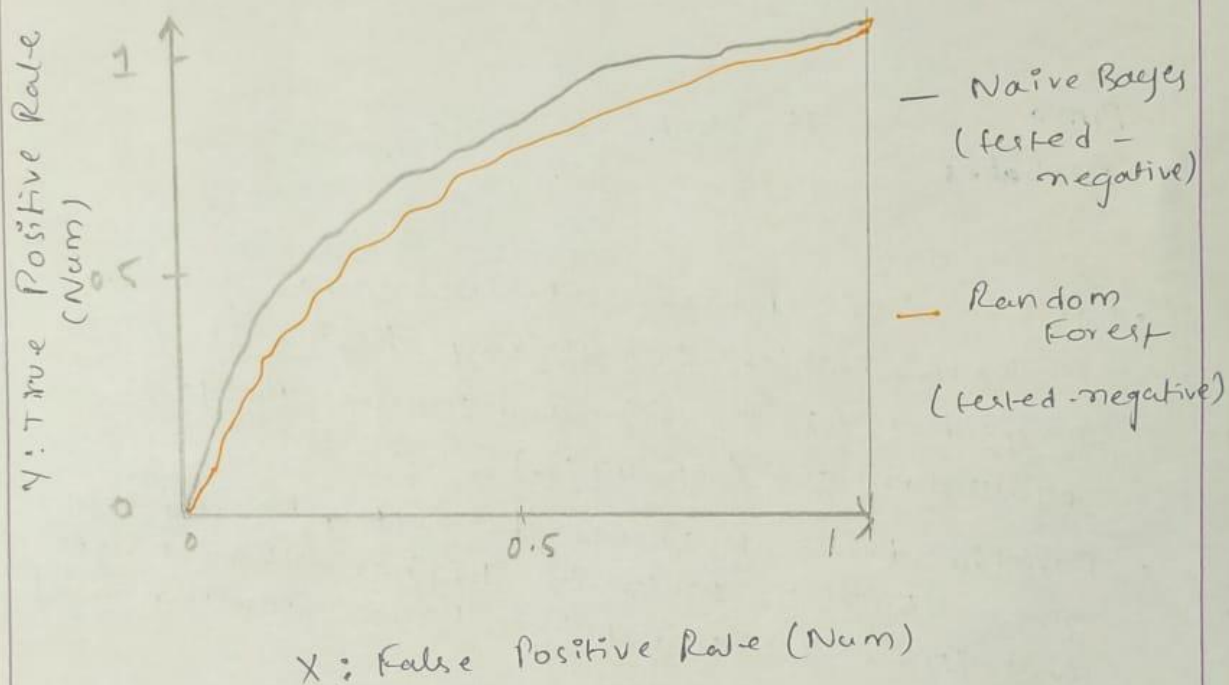
3) Naive's Bayes Multinomial

Dataset	Bayes Net	Naive Bayes	Naive Bayes Multinomial
Iris	93.20%.	95.53%.	95.47%.
Pima Diabetes	75.25%.	75.75%.	59.82%.

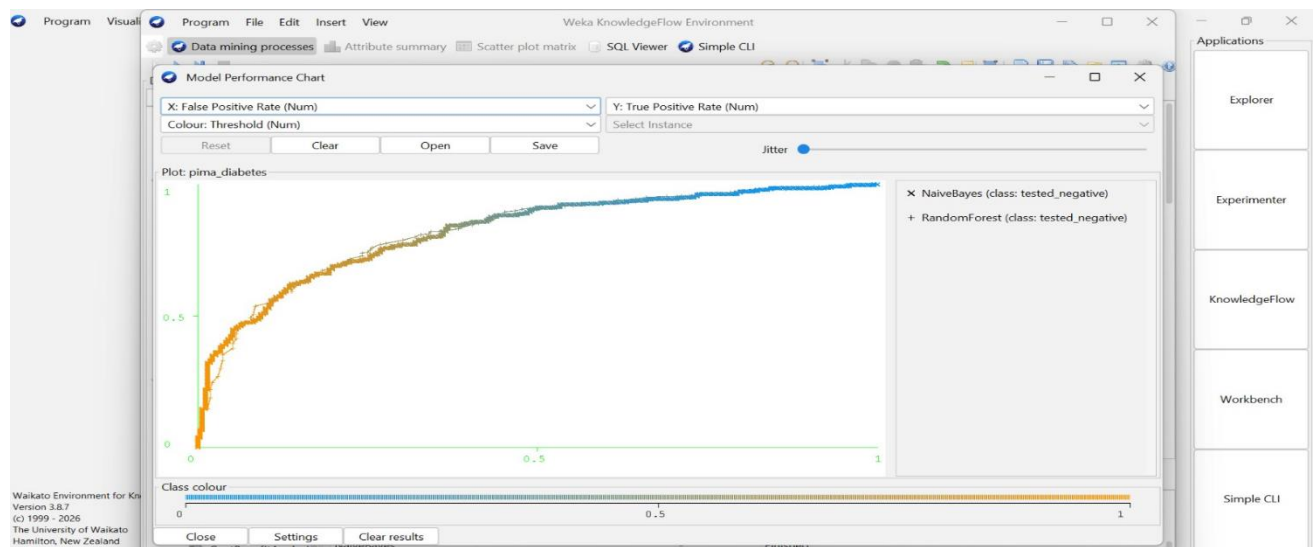
- For the Iris dataset, Naive Bayes gives 95.53% , which is slightly higher than BayesNet (93.20%.) and Naive Bayes Multinomial (95.47%.)
- For the Pima Diabetes dataset, Naive Bayes gives 75.75% . which is higher than Bayes Net (75.25%.) and Naive Bayes Multinomial (59.82%.)
- Therefore , Naive Bayes Performs best among the three classifiers for both datasets shown in the WEKA output.



ROC Curve - pima-diabetes



Alamida



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